TEACHER GUIDE

ANSWERS AND MARKS

GRADE 11 · INFORMATION AND COMMUNICATION TECHNOLOGY

UNIT 5
Web Designing Using Multimedia



! FOR THE TEACHER ONLY

This booklet has the answers to every exercise in the Unit 5 Student Workbook, in the same order. The exercise letters match the workbook.

i WHAT IS IN THIS GUIDE

- **Marks and pacing** - every exercise and its marks
- **Class marks record** - one page to copy for each class
- **Practicals at a glance** - what to check in Practicals 1 - 8, textbook links, misprints, a quiz
- **Writing spaces** - what students write in every writing space
- **Answers** - chapters 1 to 4 and the Unit 5 Test, in the order of the workbook
- **Extra** - common mistakes, homework, a practical test and the finished web page

★ BEFORE THE FIRST LESSON

- ☐ Notepad, a web browser, an HTML editor and KompoZer work on every computer.
- ☐ A shared folder has srilanka.jpg and a few small pictures.
- ☐ The projector works, so you can show each step first.

"Students remember the pages they build. Let them type, save, refresh and try again."

- THE KING OF ICT

TEACHER GUIDE

Marks and Pacing

You decide how much class time each part needs

Chapter	Exercise	Marks
Chapter 1	A True or false	8
	B Match the websites	8
	C Complete the site map	5
	D Make the media smaller	8
Chapter 2	A Element, attribute or value?	8
	B Read the colour codes	6
	C True or false	8
	D Match the formatting tags	8
	E Complete the code	6
Chapter 3	A Match the image attributes	6
	B Write the output	6
	C Complete the table code	8
	D Choose the correct answer	8
Chapter 4	A Static or dynamic?	8
	B Good domain names?	6
	C Choose the correct answer	8
End	Unit 5 Test	40
TOTAL		**155**

! SUGGESTED SHAPE OF ONE CHAPTER

READ	Teach the chapter pages. Students read with you and underline the key words.
DO	The practical tasks on real computers, in pairs.
WRITE	The written exercises. Walk around and help.
MARK	Mark the work together. Fill the tick boxes.

★ TEACHING TIPS

- Start with real websites: let the class say what kind of web application each one is (5.1).
- Ask: who are the users of our school website? Parents, students, old pupils ... Then plan for them.
- Draw a site map of the school website on the board together before anyone writes code.
- Show a page with long paragraphs and the same page with lists. Students see why lists help (5.1.4).
- Make a browser window narrow on the projector: some pages fit the screen, some do not (5.1.5).

TEACHER GUIDE

Class Marks Record

Copy this page for each class

No	Student name	Ch 1 (29)	Ch 2 (36)	Ch 3 (28)	Ch 4 (22)	Test (40)	Total (155)
1							
2							
3							
4							
5							
6							
7							
8							
9							
10							
11							
12							
13							
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17							
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19							
20							
21							
22							
23							
24							
25							
26							
27							
28							
29							

TEACHER GUIDE

Practicals at a Glance

Textbook activities on pages 132 - 166

Practical	Students do	Check that ...
1 (pp. 132, 134)	examine four websites; plan a school website in a group	they name a strength and a weakness of each site; the plan has the purpose, the users, the content, a site map, one layout and small media.
2 (Activity No 01)	myfirst.html in Notepad; coffeecup.html in an HTML editor	the file is saved as All Files (not .txt); the title shows on the title pane; the preview tool is used.
3 (pp. 142, 144)	the name of the school in **<h1>** - **<h6>** with colours, in Sinhala (iskoola pota)	Encoding: Unicode in the Save As box; every font and header tag is closed.
4 (pp. 145 - 152)	the Sri Lanka page: image, paragraph, line breaks, lists, hyperlink, table	the image path is their own folder; the link opens www.president.gov.lk; the page matches Figure 5.14.
5 (p. 156)	the table with rowspan and colspan; their team's website	Row 1 Cell 1 has rowspan="2", Row 3 Cell 1 has colspan="3".
6 (pp. 158 - 160)	Systems of the Human Body with KompoZer: 6 pages, a 6 x 3 table	the images are in one folder; the five links work; the cells are merged as in Figure 5.41.
7 (p. 162)	install XAMPP or WAMP and Joomla; build the team's website	a template is chosen and the articles show in the preview.
8 (pp. 164 - 166)	free web hosts, www.nic.lk, publish the team's website	two hosts with their web addresses; every page and link is tested.

MATERIALS AND SAFETY

- Notepad and a web browser on every computer; one HTML editor (CoffeeCup, Seamonkey Composer or Eclipse) and KompoZer installed by you before Practicals 2 and 6.
- A shared folder with srilanka.jpg and a few small pictures for Practicals 4 - 6.
- XAMPP or WAMP and Joomla (Practical 7) need the teacher's computer or a lab server - install them first.
- Only you publish a website on a real web host (Practical 8); students may use a free host with your OK.

Textbook links

Workbook	Textbook	Activities
Chapter 1 - Planning a website (5.1 - 5.5)	5.1.1 - 5.1.6, pp. 125 - 134	132, 134
Chapter 2 - HTML: your first web page (5.6 - 5.11)	5.1.7 - 5.1.10, pp. 134 - 144	No 01, 142, 144
Chapter 3 - Images, lists, links, tables (5.12 - 5.17)	5.1.10, pp. 144 - 156	147 - 156
Chapter 4 - Software, publishing (5.18 - 5.23)	5.2 - 5.3, pp. 157 - 166	162 - 166
Unit 5 in One Page	Summary, p. 167	-

TEACHING TIP

- Show a 5 MB photo and a cropped, compressed copy. Compare the file sizes and the loading time.

! TEXTBOOK NOTES - MISPRINTS (THE WORKBOOK IS CORRECT)

- p. 138: "/" is called a back slash - it is a (forward) slash. The font example says Times New Roman but its code uses face="iskoolaPota".
- p. 140: "Ariel" = Arial. p. 142: the output of ** SRI LANKA ** is printed plain, and **** is shown bold - browsers show emphasis in italics.
- pp. 145, 147: "**< img**" and "**< p align**" have a space after "<"; pp. 145 - 151: src=" C:\..." has a space inside the quotes. "section 5.10" (p. 145) = 5.1.10.
- p. 146: "Figure 5.23 - Output of code in Figure 5.24" = Figure 5.22. p. 147: " :" and " " need semicolons: ** **.
- p. 149: **<ul type = "disc"** needs its ">". p. 150: "(Figure 5.31)" = Figure 5.28; the description list ends with **<dl>** - it must be **</dl>**.
- p. 151: mixed quotes round the href value. p. 152: "more the Mouse" = move the mouse. p. 153: "Figure 36" = Figure 5.32.
- pp. 153 - 155: "PrincipleForms" / "Principle Forms of Land use" = Principal Forms of Land use. p. 154: **border = "1"width** needs a space.
- p. 157: "HLML" = HTML; "Adobe dream viewer" = Adobe Dreamweaver; "Microsoft front page" = Microsoft FrontPage. p. 159: "6 raws" = 6 rows. p. 164: "ww.nic.lk" = www.nic.lk.

? QUICK ORAL QUIZ - START OR END A LESSON WITH IT

Ask the class ...	Answer
1. What does HTML stand for?	**Hyper Text Markup Language**
2. Name a web application for e Commerce.	**www.amazon.com ...**
3. What shows the pages and the links of a website?	**A site map**
4. Name one way to reduce the size of a video.	**Reduce the frame size ...**
5. The two main parts of an HTML document?	**Head and body**
6. The three parts of a tag?	**Element, attribute, value**
7. The range of the font size? The default?	**1 - 7; 3**
8. Which tag has no closing tag?	**<img>, **
9. Which attribute merges columns?	**colspan**
10. Name an open source CMS.	**Joomla, WordPress, Drupal**
11. Who assigns domain names?	**ICANN**
12. Software that transfers a website to the host?	**FTP software**

★ TEACHING TIPS

- Talk about copyright: whose photo is it? Ask before you use it. Fair use is only for news, study, criticism.
- Most first problems are myfirst.html.txt files. Show "Save as type: All Files" on the projector.
- In File Explorer tick View, File name extensions, so students see .html at the end of the name.
- Make the class say "save, then refresh" after every change to the code.
- Play "tag ping-pong": you say a tag, the class says its closing tag and what it does.
- Keep a map of Sri Lanka (srilanka.jpg) in a shared folder before Practical 4.

WRITING SPACES

What Students Write

Read each note out or write it on the board - the students copy it into the space

Other pages

No.	Page	What the students write
1	2	**What this unit is about** Web designing: the content of a website, the fundamentals of web designing, publishing a website and maintaining a website.

Chapter 1

No.	Page	What the students write
2	4	**Content management** Content management is the systematic organisation of the selected content of a website, so that users can use it effectively.
3	5	**The size of media files** Big media files take long to download from the web server and users may leave. Reduce the size of every file.

Chapter 2

No.	Page	What the students write
4	8	**HTML** HTML (Hyper Text Markup Language) is the language of WWW documents: a markup language, not a programming language.
5	8	**Element, attribute, value** A tag has three parts: element, attribute and value. Attributes go in the opening tag; values within inverted commas.
6	9	**Head and body** An HTML document has two main parts: the head, with the title, and the body, with the content of the web page.

Chapter 3

No.	Page	What the students write
7	12	**The image tag** **<img>** is an empty tag (no closing tag). src gives the image file; alt, the text shown if the image is missing.
8	14	**Table tags** **<tr>** adds a row, **<td>** a data cell, **<th>** a bold header cell. colspan merges columns; rowspan, rows.
9	14	**Three kinds of lists** Unordered list ** **, ordered list ** ** and description list **<dl>** with **<dt>** and **<dd>**.

TEACHING TIP

- Colour codes: let students change #FF0000 to #00FF00 and #0000FF and watch the text change.

Chapter 4

No.	Page	What the students write
10	18	**Static and dynamic websites** Static: the same content, made with HTML. Dynamic: often changed; PHP, ASP.net or JSP make the pages from a database.
11	18	**Publishing a website** To publish a website: web designing software, a web host, a domain name, FTP software and an Internet connection.

ANSWERS

Chapter 1

Planning a website

A True or false

8 marks

1. **T** - Web content includes all the information and services a website gives its users.
2. **T** - Age, gender and educational level help us to understand the users of a website.
3. **T** - Content management is the systematic organisation of the selected content of a website.
4. **F** - One web page should hold as much information as possible.
5. **T** - A site map shows the web pages to be created and the hyperlinks among them.
6. **F** - Each page of one website should have a different background and different fonts.
7. **T** - A website should adjust the size of its web page to the screen automatically.
8. **F** - You may use any picture of another website without permission.

Why false: one page should hold only a limited amount of information; all the pages keep the identity of the website (the same backgrounds and fonts); you need the owner's permission.

B Match the websites

8 marks

1. www.nie.lk → E	**2.** www.amazon.com → B
3. www.youtube.com → H	**4.** www.adcash.com → G
5. www.rupavahini.lk → D	**6.** www.linkedin.com → F
7. www.bing.com → A	**8.** www.nelsonmandela.org → C

1 mark each. Every website is an example of its kind on textbook pages 125 - 126.

C Complete the site map

5 marks

1. Events	**2.** About Us	**3.** Contact	**4.** Gallery
5. Science Day	**6.** Quiz	**7.** Photos	**8.** Videos

Half a mark each. Boxes 1 and 4 have sub pages: Events (with Science Day and Quiz) and Gallery (with Photos and Videos), in either order. Boxes 2 and 3: About Us and Contact, in either order. **9.** 9 (the Home Page and eight more pages) (1 mark).

D Make the media smaller

8 marks

A way to reduce the file size	Graphics	Video	Audio
Reduce the bit rate	☐	☐	☑
Remove unnecessary parts with Crop / Trim	☑	☐	☐
Reduce the frame size	☐	☑	☐
Reduce the number of channels	☐	☐	☑
Graphic compression	☑	☐	☐
Adjust the frame rate	☐	☑	☐
Reduce the sample rate	☐	☐	☑
Remove unnecessary frames and sound	☐	☑	☐

1 mark each (textbook page 133).

ANSWERS

Chapter 2

HTML: your first web page

A Element, attribute or value?

8 marks

Part of the tag	Element	Attribute	Value
`font`	☑	☐	☐
`size`	☐	☑	☐
`"5"`	☐	☐	☑
`"verdana"`	☐	☐	☑
`color`	☐	☑	☐
`"#0000FF"`	☐	☐	☑
`p`	☑	☐	☐
`"justify"`	☐	☐	☑

1 mark each. The element is the name of the tag; the attributes are in the opening tag; the values are within inverted commas.

B Read the colour codes

6 marks

1. `#FF0000` → Red	**2.** `#000000` → Black	**3.** `#FFFFFF` → White
4. `#FFFF00` → Yellow	**5.** `#FF00FF` → Fuchsia	**6.** `#00FFFF` → Aqua

1 mark each (Figure 5.17). FF0000: all red, no green, no blue.

TEACHING TIP

- Let students change <h2> to <h1> ... <h6> in Figure 5.15 and compare the sizes on the screen.

C True or false

8 marks

1. **F** - HTML is a programming language like Pascal.
2. **T** - A web page typed in Notepad is saved as All Files, with a name that ends in .html.
3. **T** - **<TITLE>**, **<Title>** and **<title>** do the same thing.
4. **F** - The web browser shows an error message when a tag is written incorrectly.
5. **T** - In the colour code #0000FF, the last two digits are for blue.
6. **F** - Attributes are written in the closing tag.
7. **T** - The font size can be from 1 to 7; the default is size 3.
8. **F** - **<h6>** makes the biggest and the boldest letters.

Why false: HTML is a markup language; the browser ignores a wrong tag without a message; attributes go in the opening tag; **<h1>** makes the biggest letters.

D Match the formatting tags

8 marks

1. **<b>** → E	**2.** **<i>** → C	**3.** **<u>** → H	**4.** **** → F
5. **<s>** → A	**6.** **<sup>** → G	**7.** **<sub>** → D	**8.** **<marquee>** → B

E Complete the code

6 marks

1. `title`	**2.** `head`	**3.** `center`	**4.** `face`	**5.** `color`	**6.** `h2`

1 mark each. The extra words are **size** and **body**. The finished code is Figure 5.15.

★ TEACHING TIPS

- Tell students the textbook path C:\Users\Pictures\srilanka.jpg must be changed to their own folder.
- The <font> and tags of the textbook still work in browsers; newer websites use CSS for this.
- Show www.nic.lk on the projector: how a .lk domain name is registered.

ANSWERS

Chapter 3

Images, lists, links and tables

A Match the image attributes

6 marks

1. `src` → B	**2.** `alt` → E	**3.** `width` → F
4. `height` → D	**5.** `border` → A	**6.** `align` → C

1 mark each (Table 5.2).

B Write the output

6 marks

(a)

A. Paddy
B. Tea
C. Rubber

(b)

i. Sinhala
ii. Tamil
iii. English

(c)

- Kandy
- Galle
- Jaffna

2 marks each: 1 mark for the right numbering or bullets (A, B, C / i, ii, iii / squares), 1 mark for the items in order.

C Complete the table code

8 marks

1. `table`	**2.** `1`	**3.** `colspan`	**4.** `tr`
5. `td`	**6.** `rowspan`	**7.** `2`	**8.** `table`

1 mark each (Figure 5.36): colspan="2" makes LAND USE go across two columns; rowspan="2" makes 353,000 go down two rows.

D Choose the correct answer

8 marks

1. (c) .png	**2.** (b) ** **
3. (c) left	**4.** (b) keep a gap between words
5. (d) "I"	**6.** (c) **<dd>**
7. (c) href	**8.** (a) between the cells

ANSWERS

Chapter 4

Software, publishing and maintenance

A Static or dynamic?

8 marks

The website ...	Static	Dynamic
Its content is not changed for a period of time.	☑	☐
It is made with PHP, ASP.net or JSP.	☐	☑
Users are not directly connected to the web server.	☑	☐
Its server makes the pages from a database.	☐	☑
It is safer, but the coding takes more time.	☑	☐
It shows exam results for an index number.	☐	☑
Its content changes only when its code is changed.	☑	☐
It suits commercial, social and exam websites.	☐	☑

1 mark each (textbook pages 160 - 161).

B Good domain names?

6 marks

Domain name	Good	Not good
www.nie.lk (National Institute of Education)	☑	☐
www.myweb2.org	☐	☑
www.wikipedia.org (a non commercial website)	☑	☐
www.the-best-online-shop-in-sri-lanka.com	☐	☑
www.galleteashop.lk (a tea shop in Galle)	☑	☐
www.abc123.lk (the website of a school)	☐	☑

1 mark each. Not good: myweb2 has a number (it may be heard as mywebtu); a long name with lines is hard to remember and type; abc123 does not show the school and has numbers.

C Choose the correct answer

8 marks

1. (b) WYSIWYG	**2.** (d) KompoZer
3. (b) Drupal	**4.** (a) taking back ups
5. (a) ICANN	**6.** (d) FTP software
7. (b) shared host	**8.** (d) administrator

MARKING SCHEME

Unit 5 Test

Total 40 marks

1 Choose the correct answer

8 marks (1 each)

1. (b) e Commerce	**2.** (a) their age
3. (b) Crop / Trim	**4.** (b) Markup
5. (c) body	**6.** (b) 3
7. (a) a CMS	**8.** (d) Update and manage

2 Fill in the blanks

8 marks (1 each)

1. map	**2.** legal	**3.** angular	**4.** face
5. center	**6.** break	**7.** static	**8.** domain

TEACHING TIP

- Build the table of Practical 5 on the board, cell by cell: count the rows first, then the cells in each row.

3 Which tag made it?

8 marks (1 each)

1. `<b>`	**2.** `<s>`
3. `<sup>`	**4.** `<sub>`
5. `<ul> (with <li>)`	**6.** `<ol type="I"> (with <li>)`
7. `<img src="...">`	**8.** `<a href="...">`

Accept **`<strike>`** for 2 and **`<li>`** for 5 and 6; for 8 accept "anchor tag".

4 Answer these

16 marks

1. Name two kinds of web applications and give one example website of each. (2 marks)

Any two kinds with a correct example (1 mark each), e.g. e Commerce - www.amazon.com; News - www.bbc.co.uk; Search engine - www.google.com; Social media - www.facebook.com.

2. Write two ways to reduce the size of a video file for a website. (2 marks)

Any two (1 mark each): reduce the frame size; reduce the resolution; adjust the frame rate; remove unnecessary frames and sound.

3. Write the HTML code to show the text SRI LANKA in red, in size 5, with the face Arial. (3 marks)

`<font size="5" face="Arial" color="red">SRI LANKA</font>`: 1 mark for **`<font`** ... **`</font>`**, 1 mark for size and face, 1 mark for color (also accept "#FF0000"; any order).

4. Write the HTML code for a list that shows A. Tea B. Rubber (3 marks)

`<ol type="A"> <li>Tea</li> <li>Rubber</li> </ol>`: 1 mark for **`<ol>`** ... **`</ol>`**, 1 mark for type="A", 1 mark for the two **`<li>`** items.

5. Write two differences between a static website and a dynamic website. (2 marks)

Any two (1 mark each): static - content not changed, dynamic - often changed; static - made with HTML, dynamic - PHP, ASP.net, JSP with a database; static - users not directly connected to the web server, dynamic - directly connected; static - safer, dynamic - security is threatened more.

6. Write four things a web publisher needs to publish a website. (2 marks)

Any four, half a mark each: a web designing software; a web host; a domain name; FTP; an Internet connection.

7. Write two things to consider when you select a domain name. (2 marks)

Any two (1 mark each): it shows the institute or the objective; a suitable top level domain (.com, .edu, .org, .lk); it is short; no numbers or lines.

TIP: Keep a printed copy of the finished srilanka.html on the wall during Chapter 3.

WHAT STUDENTS USUALLY GET WRONG

- They save **myfirst.html.txt**: Save as type must be **All Files**.
- They forget the "/" of a closing tag, or close the tags in the wrong order: **<h2> ... </h2>**.
- They write an attribute in the closing tag, or a value without inverted commas.
- They give the font a size of 8 or more. The size is **1 to 7**; the default is 3.
- They mix up colspan (columns, across) and rowspan (rows, down).
- They mix up **<td>** (a cell) and **<tr>** (a row), or **<dt>** and **<dd>**.
- They think a static website changes by itself. Only the designer changes its HTML code.
- They mix up the web host (keeps the website) and the domain name (its name).

HOMEWORK AND EXTRA PRACTICE

- Find one example website for each kind of web application and say who its users are.
- Draw a site map of the website of a club or society of your school.
- On paper, write the HTML of a page about your village with a heading, a picture, a list and a table.
- Fast learners: link three of your pages together and add a table with colspan and rowspan.

PRACTICAL TEST - 30 MINUTES, 20 MARKS (OPTIONAL)

Use it after the unit, in the computer room. Write the steps on the board. Mark on the screen.

Step	The student must ...	Marks
1	Open Notepad and type the **<html>**, **<head>**, **<title>** and **<body>** tags.	3
2	Add the title text **MY SCHOOL** with **<h1>**, ** ** and a font with a face and a colour.	3
3	Add a picture with **<img>** (src, width, alt).	2
4	Add a paragraph about the school, justified.	2
5	Add an ordered list (type "A") of three subjects.	3
6	Add a table of 3 rows with a caption and **<th>** headings.	4
7	Save it as **myschool.html** (All Files) and open it in the browser.	3
	Total	**20**

TEACHING TIPS

- colspan goes across (columns), rowspan goes down (rows). Students can show it with their arms.
- Use the workbook word list as a quick quiz: read a meaning, the class says the word.

THE FINISHED srilanka.html (PRACTICAL 4) - TO CHECK ON THE SCREEN

srilanka.html

```
<html>
<head><title> Sri Lanka </title></head>
<body><h2><center><font face="arial" color="blue"> SRI LANKA </font></center></h2>
<center><img src="C:\Users\Pictures\srilanka.jpg"></center>
<center><font face="arial" size="2"> A map of Sri lanka </font></center>
<p> Sri Lanka is an island located off the southern coast of India. Sri Lanka is surrounded
by the Indian Ocean. The geography of Sri Lanka includes coastal plains in the
north and hills and mountains in the interior. The government system is a republic.
The chief of state and head of government is the
<a href="http://www.president.gov.lk" >President.</a></p>
People in Sri Lanka have four main categories in ethnically. They are <br/>
Sinhala <br/>Tamil <br/> Muslim <br/> Burghers <br/>
<table border="1" >
    <caption> Principal Forms of Land use</caption>
    <tr> <th>USE</th> <th>EXTENT(hectares)</th> </tr>
    <tr> <td>Paddy</td> <td>500,000</td> </tr>
    <tr> <td>Tea</td> <td>190,000</td> </tr>
    <tr> <td>Rubber</td> <td>163,000</td> </tr>
</table>
</body>
</html>
```

TEACHING TIPS

- If the Internet is slow, practise KompoZer offline; the Joomla practical needs XAMPP or WAMP first.
- Assign the four roles (author, editor, publisher, administrator) to groups for the class website.
- Ask the class to say four domain names aloud; the others write them. Numbers and dashes cause mistakes.
- Revise the seven stages as a cycle on the board: concept, design develop, domain name, hosting, upload, test, update and manage.
- Read one writing-space note at the start of each lesson: the class writes it while you set up.
- O/L papers often ask for HTML code. Let students write code on paper first, then type and test it.
- Mark the unit test together the next day. Students learn most from the questions they got wrong.
- Fast learners can add a hyperlink to the map picture and a second page linked back to the first.
- Give each group a real website to judge with the five points of planning: purpose, users, content, management, layout.

TIP: Let students open the page source of a simple website (Ctrl + U) and find the head and the body.